

Moment-generating Functions

Discrete Distributions (Poisson, Binomial, Geometric)

Poisson Distribution

$$\begin{aligned}
 M(t) &= E(e^{tx}) = \sum_{k=0}^{\infty} e^{tk} \frac{\lambda^k e^{-\lambda}}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{e^{tk} \lambda^k}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{(e^t \lambda)^k}{k!} = \\
 &\quad \boxed{e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}} \\
 &= e^{-\lambda} \times e^{e^t \lambda} = e^{-\lambda + e^t \lambda} = e^{\lambda(e^t - 1)} \\
 M(t) &= e^{\lambda(e^t - 1)} \\
 M'(t) &= \lambda e^t e^{\lambda(e^t - 1)} \\
 M''(t) &= \lambda e^t e^{\lambda(e^t - 1)} + \lambda^2 e^{2t} e^{\lambda(e^t - 1)}
 \end{aligned}$$

Binomial Distribution

$$\begin{aligned}
 M(t) &= E(e^{tx}) = \sum_{k=0}^n e^{tk} \binom{n}{k} p^k (1-p)^{n-k} = \sum_{k=0}^n \binom{n}{k} (pe^t)^k (1-p)^{n-k} = \\
 &\quad \boxed{(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} \quad \text{where} \quad a = pe^t \quad b = (1-p)} \\
 &= (pe^t + (1-p))^n = (1-p + pe^t)^n \\
 M(t) &= (1-p + pe^t)^n \\
 M'(t) &= npe^t (1-p + pe^t)^{n-1} \\
 M''(t) &= npe^t (1-p + pe^t)^{n-1} + np^2 e^{2t} (n-1)(1-p + pe^t)^{n-2}
 \end{aligned}$$

Geometric Distribution

$$\begin{aligned}
 M(t) &= E(e^{tx}) = \sum_{k=1}^{\infty} e^{tk} (1-p)^{k-1} p = p \sum_{k=1}^{\infty} e^{tk} (1-p)^{k-1} = \\
 &\quad \boxed{e^{tk} = e^{t(k-1+1)} = e^{t(k-1)} e^t} \\
 &= pe^t \sum_{k=1}^{\infty} e^{t(k-1)} (1-p)^{k-1} = pe^t \sum_{k=1}^{\infty} (e^t (1-p))^{k-1} = pe^t \sum_{n=0}^{\infty} (e^t (1-p))^n = \\
 &\quad \boxed{\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}, \quad |r| < 1, \quad \text{where} \quad r = (1-p)e^t} = \frac{pe^t}{1 - (1-p)e^t} \\
 M(t) &= \frac{pe^t}{1 - (1-p)e^t} \\
 M'(t) &= \frac{pe^t}{(1 - (1-p)e^t)^2} \\
 M''(t) &= \frac{pe^t (1 + (1-p)e^t)}{(1 - (1-p)e^t)^3}
 \end{aligned}$$

Continuous Distributions (Uniform, Exponential, Normal)

Uniform Distribution

$$\begin{aligned}
 M(t) &= E(e^{tx}) = \int_a^b \frac{e^{tx}}{b-a} dx = \frac{1}{b-a} \int_a^b e^{tx} dx = \frac{e^{tx}}{t(b-a)} \Big|_a^b = \frac{e^{bt} - e^{at}}{t(b-a)} \\
 M(t) &= \frac{e^{bt} - e^{at}}{t(b-a)} \\
 M'(t) &= \frac{(be^{bt} - ae^{at})t - (e^{bt} - e^{at})}{t^2(b-a)} \\
 M''(t) &= \frac{(b^2 e^{bt} - a^2 e^{at})t^2 - 2(be^{bt} - ae^{at})t + 2(e^{bt} - e^{at})}{t^3(b-a)}
 \end{aligned}$$

Exponential Distribution

$$\begin{aligned}
 M(t) &= E(e^{tx}) = \int_0^{\infty} e^{tx} \times \lambda e^{-\lambda x} dx = \lambda \int_0^{\infty} e^{tx} e^{-\lambda x} dx = \lambda \int_0^{\infty} e^{(t-\lambda)x} dx = \frac{\lambda e^{(t-\lambda)x}}{t-\lambda} \Big|_0^{\infty} = \frac{\lambda \times 0}{t-\lambda} - \frac{\lambda \times 1}{t-\lambda} = -\frac{\lambda}{t-\lambda} = \frac{\lambda}{\lambda-t}, \quad t < \lambda \\
 M(t) &= \frac{\lambda}{\lambda-t} \\
 M'(t) &= \frac{\lambda}{(\lambda-t)^2} \\
 M''(t) &= \frac{2\lambda}{(\lambda-t)^3}
 \end{aligned}$$

Normal Distribution

$$M(t) = E(e^{tX}) = \int_{-\infty}^{\infty} e^{tx} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(tx - \frac{(x-\mu)^2}{2\sigma^2}\right) dx = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(-\frac{(x - (\mu + \sigma^2 t))^2}{2\sigma^2}\right) + \mu t + \frac{\sigma^2 t^2}{2} dx = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)$$

$$\begin{aligned}
 M(t) &= \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right) \\
 M'(t) &= (\mu + \sigma^2 t) \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right) \\
 M''(t) &= (\sigma^2 + (\mu + \sigma^2 t)^2) \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)
 \end{aligned}$$

note : $e^x = \exp(x)$

@salyamq2